

DPS, IND

Subject: maths | Class: Grade 8

Time Allowed: 45 minutes

Maximum Marks: 25

All questions are compulsory unless stated otherwise.

Name: _____

Roll Number: _____

Section: _____

Section A: Multiple Choice Questions

Choose the correct option

1. [Easy] What is the value of x in the equation $2x + 5 = 11$? (A) 2, (B) 3, (C) 4, (D) 5 [1 Mark]
2. [Moderate] In a right-angled triangle, the length of the hypotenuse is 10 cm and one of the other sides is 6 cm. What is the length of the third side? (A) 4 cm, (B) 6 cm, (C) 8 cm, (D) 12 cm [1 Mark]
3. [Easy] If a car travels 250 km in 5 hours, what is its average speed in km/h? (A) 40 km/h, (B) 50 km/h, (C) 60 km/h, (D) 70 km/h [1 Mark]
4. [Moderate] A rectangular garden measures 12 m by 8 m. What is its area in square meters? (A) 20 m^2 , (B) 40 m^2 , (C) 60 m^2 , (D) 80 m^2 [1 Mark]
5. [Challenging] What is the value of $\sin(60^\circ)$? (A) 0.5, (B) 0.6, (C) 0.7, (D) 0.8 [1 Mark]

Section B: Short Questions

Answer in one or two sentences

6. [Moderate] What is the formula for the area of a circle? [1 Mark]
7. [Moderate] What is the difference between a rational number and an irrational number? [1 Mark]
8. [Challenging] What is the concept of integration in mathematics? [1 Mark]

9. [Moderate] What is the formula for the volume of a rectangular prism? [1 Mark]

10. [Challenging] What is the concept of a function in mathematics? [1 Mark]

Section C: Long Questions

Answer in 2-3 paragraphs

11. [Challenging] Explain the concept of integration and its importance in real-life applications. [1 Mark]

12. [Moderate] Solve the equation $2x^2 + 5x - 3 = 0$ using the quadratic formula. [1 Mark]

13. [Challenging] Find the area under the curve $y = 2x^2 + 3x - 1$ between $x = 0$ and $x = 2$. [1 Mark]

14. [Challenging] Explain the concept of a derivative in mathematics and its importance in physics. [1 Mark]

15. [Moderate] Solve the inequality $2x - 5 > 3$. [1 Mark]

Section D: Diagram/Graph-Based Questions

Draw a diagram or graph to illustrate your answer

16. [Easy] Draw a histogram to represent the following data: 5, 7, 9, 11, 13. [1 Mark]

17. [Easy] Draw a line graph to represent the following data: 2, 4, 6, 8, 10. [1 Mark]

18. [Easy] Draw a bar graph to represent the following data: 10, 20, 30, 40, 50. [1 Mark]

19. [Moderate] Draw a scatter plot to represent the following data: (2, 4), (4, 6), (6, 8), (8, 10). [1 Mark]

20. [Moderate] Draw a circle to represent the equation $x^2 + y^2 = 16$. [1 Mark]

Section E: Numerical Problems

Show your working and calculations

21. [Easy] Solve the equation $2x + 5 = 11$. **[1 Mark]**

22. [Moderate] Find the value of x in the equation $x/2 + 3 = 7$. **[1 Mark]**

23. [Moderate] Solve the inequality $2x - 5 > 3$. **[1 Mark]**

24. [Easy] Find the value of y in the equation $y/4 = 9$. **[1 Mark]**

25. [Challenging] Solve the equation $x^2 - 4x + 4 = 0$. **[1 Mark]**

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